A dual-channel data-generating process for biomarker-treatment interactions in aggregated N-of-1 trials: two parameterizations of one conditional-mean slope

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Contents

Abstract **2**

1 Introduction **3**

2 The dual-channel architecture **4**

2.1 Definition 4

2.2 One Gaussian law, one conditional-mean slope 5

2.3 Relation to mixture and location-scale models 5

3 Analytical results **6**

3.1 The slope profile over all visits 6

3.2 The fitted interaction coefficient 6

3.3 Identifiability 7

3.4 Noncentrality and the scale of additivity 7

4 Simulation study **8**

4.1 Aims 8

4.2 Data-generating mechanisms 8

4.3 Estimand 9

4.4 Methods 9

4.5 Performance measures 9

4.6 Validation gate 9

4.7 Test calibration under the null 10

4.8 Power over the weight grid . . . 10
4.9 The fitted coefficient . . . 12
4.10 Additivity of power . . . 13
4.11 The equal-total line and carryover . . . 14
4.12 The G9 analysis and the pharmacological half-life . . . 15

5 When to use the dual-channel architecture **17**

6 Discussion **18**

Glossary **19**

Abbreviations **20**

Conflict of interest **20**

Funding **20**

Reproducibility **20**

References **21**

Abstract

Background. Simulation studies of a predictive biomarker in aggregated N-of-1 trials must state how the biomarker-treatment interaction enters the data-generating process (DGP). Two encodings are in use. The mean-shift encoding adds a biomarker-proportional shift to the drug-response component on drug. The correlation encoding makes the biomarker-response correlation present on drug and absent off drug. We study a DGP with independent weights on both.

Methods. We derived the joint law of the dual-channel DGP, the conditional slope of the outcome on the biomarker at every visit, and the population value of the fitted interaction coefficient. We then simulated a 3×3 grid of channel weights in three designs and four carryover half-lives, at a total of 70 participants, with 1000 replicates per cell. Common random numbers paired every weight cell within a replicate. A validation gate required the boundary cells to reproduce the single-encoding DGPs draw for draw.

Results. The boundary cells reproduced the single-encoding DGPs exactly in 1,800 paired comparisons. At zero half-life the fitted coefficient was additive in the two weights. Combined power exceeded the probability-scale additive benchmark by 0.22 to 0.34. Doubling one weight produced the same excess, 0.21 to 0.35. A benchmark that adds the two contributions

on the probit scale, with the critical value calibrated to the observed size, matched every combined cell within two MCSE. At a half-life of 1 week and weight 0.45, Hybrid power was 0.818 (MCSE 0.012) under the mean-shift encoding. It was 0.723 (MCSE 0.014) under the correlation encoding. The mean-shift estimate exceeded its nominal value by up to 20 percent.

Conclusions. Under a normally distributed biomarker, the two channels act on the same quantity, the conditional mean slope. They are two parameterizations of one Gaussian model, not two mechanisms. They differ only in how much interaction persists off drug and in a small reduction of on-drug residual variance. The dual-channel DGP is useful as a two-parameter family of off-drug interaction profiles. A genuinely distinct second channel would require a variance that depends on the biomarker or a responder mixture.

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1 Introduction

Hendrickson et al.[1] supplied the design machinery on which this series builds. That work introduced the Hybrid design, the decomposition of the outcome into three response components, and the correlation encoding of the biomarker-treatment interaction. A companion paper extends that DGP and describes the extensions[2]. We call the result the extended DGP and the DGP of Hendrickson et al. the original DGP.

A simulation study of power for a predictive biomarker has to say how the interaction is generated. The usual convention in trial simulation is the mean-shift encoding. The treatment effect is scaled by the biomarker in the conditional mean. The original DGP uses the correlation encoding instead. The interaction is a correlation between the biomarker and the drug-response component that is present on drug and absent off drug. The moderator framework of Kraemer et al.[3] and the PATH statement[4] describe the estimand. Neither fixes the generating mechanism.

This paper studies a DGP with both encodings active at once, each with its own weight. We call it the dual-channel architecture. Either encoding is recovered by setting the other weight to zero. An earlier draft of this paper claimed two results. The first was that the two channels combine super-additively in power. The second was that the two channels are distinct mechanisms that act on different moments of the response. Neither claim holds. This version replaces both.

The paper makes four contributions. First, we show that under a normally distributed biomarker the dual-channel DGP is one multivariate normal law. Both channels act on the conditional mean slope of the drug-response component on the biomarker. Neither makes the conditional variance depend on the biomarker (Section 2). Second, we derive the slope profile at every visit, off-drug visits included, and its projection onto the fitted interaction coefficient (Section 3). The two weights enter that coefficient additively, with coefficients that are equal only when there is no carryover. Third, we show that the excess of combined power over

the probability-scale additive benchmark is a property of the power curve. A single channel whose weight is doubled shows the same excess (Section 4). Fourth, we report a corrected simulation under the extended DGP. It passes a validation gate that the earlier run lacked.

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2 The dual-channel architecture

2.1 Definition

Let B_i be the biomarker of participant i, with population mean μ_{bm} and standard deviation σ_{bm}. Write $b_i = (B_i - \mu_{bm})/\sigma_{bm}$ for the biomarker standardized at its population values. The outcome at visit t is $Y_{it} = \mathrm{BL}_i - TV_{it} - PB_{it} - BR_{it}$. Here BL_i is the baseline level, TV_{it} the natural-history component, PB_{it} the placebo-response component and BR_{it} the drug-response component. The three components are reductions in symptom severity, so an increase in any of them lowers Y_{it}. Interaction coefficients are therefore negative, and the channel weights are positive (NOTATION.md, rule 5).

The extended DGP draws $(B_i, \mathrm{BL}_i, TV_{i\cdot}, PB_{i\cdot}, BR_{i\cdot})$ from one multivariate normal law. Each component has the same first-order autoregressive correlation, $\rho = 0.7$ per week. Two components correlate at $c_1 \rho^{|w_t - w_s|}$ with $c_1 = 0.2$, where w_t is the calendar week of visit t. The mean of BR_{it} follows a modified Gompertz curve in the time on drug. Off drug, it decays from the last on-drug mean at the carryover half-life $t_{1/2}$. A covariance matrix that is not positive definite stops the simulation.

The dual-channel architecture adds two weights to this law.

- The **correlation channel**, weight $c_{bm,B}$, sets $\mathrm{Cor}(B_i, BR_{it}) = c_{bm,B}\, g_t$. The profile g_t is 1 on drug, $\phi_t = 2^{-t_{sd}/t_{1/2}}$ off drug after first exposure, and 0 before first exposure. Here t_{sd} is the time since discontinuation.
- The **mean-shift channel**, weight $c_{bm,A}$, adds $c_{bm,A}\, b_i\, \sigma_{BR}\, D_{it}$ to BR_{it} after the draw. Here D_{it} is 1 on drug and 0 off drug, and σ_{BR} is the standard deviation of the drug-response component.

The two weights are calibrated to a common scale (NOTATION.md, rule 2). One unit of either weight moves the on-drug conditional mean of BR_{it} by σ_{BR} per standard deviation of the biomarker. The code labels are `combined` for the dual-channel architecture, `mean_moderation` for the mean-shift encoding alone and `mvn` for the correlation encoding alone. The weights are `c.bm_a` and `c.bm_b`.

2.2 One Gaussian law, one conditional-mean slope

The mean shift is linear in b_i, and b_i is normal. The dual-channel DGP is therefore a single multivariate normal law. Its conditional moments given the biomarker follow from standard regression algebra. At visit t,

$$\mathbb{E}[BR_{it} \mid b_i] = \mu_{BR,t} + (c_{bm,A}\, D_{it} + c_{bm,B}\, g_t)\, \sigma_{BR}\, b_i, \tag{1}$$

$$\mathrm{Var}[BR_{it} \mid b_i] = \sigma^2_{BR}(1 - c^2_{bm,B}\, g_t^2). \tag{2}$$

Equation (1) shows that both channels act on the conditional mean. Equation (2) shows that only the correlation channel changes the conditional variance, and that it lowers it. The mean shift is a fixed function of b_i, so it adds no variance once the biomarker is given. Its contribution $c^2_{bm,A}\sigma^2_{BR}$ to the marginal variance of BR_{it} disappears on conditioning. Neither conditional variance depends on b_i. The package source states the on-drug equivalence of the two channels in a comment (`R/generateData.R`, lines 189 to 195).

At the parameters used here, $\sigma_{BR} = 8$ points. The on-drug conditional variance of BR_{it} falls from 64 to 51.0 at $c_{bm,B} = 0.45$. The outcome sums four components, so the effect on $\mathrm{Var}(Y_{it} \mid b_i)$ is smaller. At the on-drug Hybrid visits the ratio is 0.978 to 0.982.

Three consequences follow.

1. On drug, the two channels are observationally the same. A pair $(c_{bm,A}, c_{bm,B})$ gives the on-drug slope $(c_{bm,A} + c_{bm,B})\, \sigma_{BR}$ and differs from another pair with the same sum only through the small variance reduction above.
2. Off drug, they differ. The mean shift is applied at on-drug visits only, so the mean-shift channel's interaction has no carryover. The correlation channel's interaction decays with ϕ_t. The drug-response mean itself carries over under both.
3. Setting $c_{bm,B} = 0$ gives the mean-shift encoding exactly. Setting $c_{bm,A} = 0$ gives the correlation encoding exactly. The simulation uses this as a validation gate (Section 4.6).

Only $c_{bm,B}$ is bounded by positive definiteness. The weight $c_{bm,A}$ has no feasibility bound. At the parameters used here, every matrix with $c_{bm,B} \leq 0.45$ was positive definite in every design and half-life. The smallest eigenvalue over all 108 design, half-life and weight combinations was 2.87.

2.3 Relation to mixture and location-scale models

An earlier draft described the two channels as distinct biological mechanisms. It called the mean shift "pharmacodynamic scaling" and the correlation "co-regulatory" activity. It also said the correlation channel carries information through the class-conditional dispersion $\mathrm{Var}[BR \mid B]$. Section 2.2 shows that, as specified, both channels move the conditional mean and neither makes the dispersion depend on the biomarker. The two descriptions are two readings of one model. They are not separable mechanisms within it.

The earlier draft also called the combined form the mean- and covariance-structure model of Arminger et al.[5]. That paper concerns finite mixtures of conditional mean- and covariance-structure models. Verbeke and Lesaffre[6] study a mixture distribution for the random effects of a linear mixed model. The dual-channel DGP has no mixture component, so neither paper describes it. Both describe the kind of extension that would create a second channel. A responder mixture with a biomarker-dependent class probability is one such extension. The latent-class companion paper develops it[7]. A mixed-effects location-scale model, in which a covariate enters the within-participant variance, is another[8]. Under either, a drug could change the spread of response as well as its mean.

The variance-ratio literature tests that second moment directly. Under a constant treatment effect, treated and control arms have the same outcome variance[9]. Reviews of parallel-group trials[10], antipsychotic trials[11] and antidepressant trials[12] found treated-arm variances equal to or smaller than control-arm variances. The correlation encoding predicts exactly that pattern. It lowers the on-drug conditional variance and leaves the marginal variance unchanged. The mean-shift encoding raises the marginal on-drug variance by $c_{bm,A}^2 \sigma_{BR}^2$. Neither encoding produces a biomarker-dependent variance.

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3 Analytical results

3.1 The slope profile over all visits

Write the outcome vector of a participant on path p as $\mathbf{Y}_i = (\mathrm{BL}_i, Y_{i1}, \ldots, Y_{iT})^\top$. The baseline visit is included, because the analysis model includes it. Conditional on the biomarker,

$$\mathbb{E}[\mathbf{Y}_i \mid B_i] = \mathbf{m}_p + \mathbf{s}_p\,(B_i - \mu_{bm}), \qquad s_{pt} = -(c_{bm,A}\,D_{pt} + c_{bm,B}\,g_{pt})\frac{\sigma_{BR}}{\sigma_{bm}}, \tag{3}$$

with $s_{p0} = 0$ at baseline. The conditional covariance $\mathbf{V}_p = \mathrm{Var}(\mathbf{Y}_i \mid B_i)$ does not depend on B_i or on $c_{bm,A}$. Both follow from the covariance matrix the DGP builds. Script `analysis/scripts/dgp-combined/03-analytic-projection.R` computes them.

3.2 The fitted interaction coefficient

The analysis fits $Y_{it} = \beta_0 + \beta_B B_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_{bm:D} B_i D_{bc,it} + u_i + \varepsilon_{it}$, where $D_{bc,it}$ is the exposure-decayed drug indicator at the true half-life (Section 4.4). The biomarker enters through β_B and $\beta_{bm:D}$. Their population values are the projection of the slope profile $\mathbf{s}_p$ onto the columns $(1, D_{bc,pt})$, pooled over paths. With generalized least squares under the true conditional covariance,

$$\begin{pmatrix} \beta_B \\ \beta_{bm:D} \end{pmatrix} = \Big(\sum_p n_p\,\mathbf{X}_p^\top \mathbf{V}_p^{-1} \mathbf{X}_p\Big)^{-1} \sum_p n_p\,\mathbf{X}_p^\top \mathbf{V}_p^{-1}\,\mathbf{s}_p, \qquad \mathbf{X}_p = (\mathbf{1}, \mathbf{D}_{bc,p}), \tag{4}$$

where n_p is the number of participants on path p.

Two cases follow from Equation (3).

- *The correlation channel.* At the true half-life, g_{pt} equals $D_{bc,pt}$ at every visit. Its slope profile lies in the span of $\mathbf{X}_p$, so the projection is exact for any weighting. Its contribution to $\beta_{bm:D}$ is $-c_{bm,B}\,\sigma_{BR}/\sigma_{bm}$ in every design and at every half-life. At the parameters used here that is -0.2344 at $c_{bm,B} = 0.45$.
- *The mean-shift channel.* Its profile is D_{pt}, which is 0 at off-drug visits where $D_{bc,pt}$ is positive. At $t_{1/2} = 0$ the two coincide and the projection is again exact. At $t_{1/2} > 0$ the profile is not in the span of $\mathbf{X}_p$. The projection then depends on the weighting and exceeds the nominal value in absolute size. The analysis model reads a step-shaped interaction through an exposure-decayed regressor.

The coefficient is therefore linear in the two weights,

$$\beta_{bm:D} = c_{bm,A}\,\kappa_A + c_{bm,B}\,\kappa_B, \qquad \kappa_B = -\sigma_{BR}/\sigma_{bm}, \tag{5}$$

with $\kappa_A = \kappa_B$ only at $t_{1/2} = 0$. Under the true conditional covariance, the mean-shift channel's coefficient exceeds the nominal value by 1 percent in the crossover design, 8 percent in the Hybrid design and 7 percent in the OL+BDC design at $t_{1/2} = 1$ week. The departure from exact linearity caused by the dependence of $\mathbf{V}_p$ on $c_{bm,B}$ was at most 9.2e-06 over the grid. The fitted model uses an estimated random-intercept and continuous AR(1) working covariance, not $\mathbf{V}_p$, so its mean-shift overshoot can differ. Section 4.9 reports it.

3.3 Identifiability

At $t_{1/2} = 0$, any two weight pairs with the same sum give the same fitted coefficient. The two weights are not separately identifiable from it. At $t_{1/2} > 0$ they enter with different coefficients, but one coefficient still cannot identify two weights. The data carry the difference in the off-drug visits. A model that estimates the off-drug slope separately could identify the off-drug profile, and with it the split between the channels. An example is a specification with a biomarker-by-lagged-exposure term. We did not fit such a model.

3.4 Noncentrality and the scale of additivity

The GLS standard error of $\beta_{bm:D}$ under the true conditional covariance is $\{[(\sigma^2_{bm}\sum_p n_p \mathbf{X}_p^\top \mathbf{V}_p^{-1}\mathbf{X}_p)^{-1}]_{22}\}^{1/2}$. The noncentrality of the test is $\Lambda = |\beta_{bm:D}|/\mathrm{SE}$. By Equation (5), Λ is nearly linear in the two weights. It is not exactly linear, because $\mathbf{V}_p$ depends on $c_{bm,B}$. Power $\pi(\Lambda) \approx \Phi(\Lambda - z_{0.975})$ is not linear in Λ. It is convex below $\pi = 0.5$ and concave above. A probability-scale benchmark $p(a,0) + p(0,b) - p(0,0)$ therefore understates the combined power whenever the combined noncentrality lies in the convex region. It overstates it in the concave region, and it can exceed 1. This holds for any two additive contributions to Λ. It holds equally when

one channel's weight is doubled. The appropriate benchmark adds the two contributions on the noncentrality scale. Section 4.10 applies it.

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4 Simulation study

We report the study under the ADEMP framework[13]. The study is exploratory.

4.1 Aims

The aims were (1) to check that the boundary cells reproduce the single-encoding DGPs, (2) to describe power over the weight grid under the extended DGP, (3) to test additivity on the scale of the fitted coefficient and of the noncentrality, and (4) to compare the equal-total weight pairs, which differ only in their off-drug profile.

4.2 Data-generating mechanisms

We used the extended DGP of Section 2 with the prazosin-PTSD calibration of the companion paper. The drug-response standard deviation was $\sigma_{BR} = 8$ points and the biomarker standard deviation $\sigma_{bm} = 15.36$ units. The factors were the following.

- Channel weights $c_{bm,A}, c_{bm,B} \in \{0, 0.225, 0.45\}$, nine cells. The anti-diagonal holds the three equal-total pairs with $c_{bm,A} + c_{bm,B} = 0.45$. They are the points $\omega \in \{0, 0.5, 1\}$ of the mixing parameterization $c_{bm,A} = (1 - \omega)\, c$, $c_{bm,B} = \omega\, c$ at a total weight $c = 0.45$.
- Three designs. The crossover design (CO) has two periods and two randomization paths. Open-label titration followed by blinded discontinuation (OL+BDC) has two paths. The Hybrid design (open-label titration, blinded discontinuation, then a brief crossover) has four paths. Each design has eight post-baseline visits.
- Carryover half-life $t_{1/2} \in \{0, 0.5, 1\}$ week. The two positive values are behavioral scenarios. The pharmacological range of prazosin is 0.1 to 0.2 week, so a separate pass of the same driver added $t_{1/2} = 0.2$ week (Section 4.12).
- A total of $N = 70$ participants, allocated across the paths of each design (35 per path in CO and OL+BDC, 18, 18, 17 and 17 in Hybrid).

Each replicate set one seed, determined by the design, half-life and replicate index, before generating the data of every weight cell. Since the Cholesky factor changes with $c_{bm,B}$ but the normal deviates do not, the nine cells of a replicate share their random numbers. Contrasts between weight cells are paired. Designs and half-lives use independent streams. Every simulated covariance matrix was positive definite, so no replicate was refused.

4.3 Estimand

The estimand is the biomarker-by-exposure coefficient $\beta_{bm:D}$ of the analysis model, in outcome points per biomarker unit. Its population value is given by Equation (5). Power is the probability of rejecting $\beta_{bm:D} = 0$ at two-sided $\alpha = 0.05$.

4.4 Methods

Each simulated trial was analyzed with the following model and two standard errors.

- **E2.** `Sx ~ bm + t + Dbc + bm:Dbc` fitted by `nlme::lme` with REML, a random intercept per participant and a continuous AR(1) residual correlation, `corCAR1(form = ~ t | ptID)`. $D_{bc,it}$ is built at the true half-life. It is 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure and 0 before first exposure. The test uses the model-based standard error and the `lme` t distribution.
- **E2cr2.** The same fit, with the CR2 cluster-robust standard error[14] and a Satterthwaite t test[15], from `clubSandwich`. Paper 02 labels it G8[16].

The analysis data include the baseline visit, with $t = 0$, $D_{bc} = 0$ and outcome BL_i, followed by the eight post-baseline visits. A fit that failed was recorded as not converged. No fallback model was fitted.

The program's current recommendation for the interaction is G9, the E2 form with the half-life chosen by AIC and a CR2 test, or F1 at a fixed analysis half-life[16]. G9 costs about eight E2 fits per trial. We fitted it to a subset of the trials, the first 200 replicates of four cells per design and half-life, and compared it with E2cr2 on the same trials (Section 4.12). We did not fit F1. Its expectancy term is not identifiable in the CO design, whose design expectancy is constant.

4.5 Performance measures

For every cell we report the rejection rate $\hat{\pi}$ with MCSE $= \{\hat{\pi}(1-\hat{\pi})/n_{\mathrm{reps}}\}^{1/2}$, the mean of the estimate, its empirical standard deviation, and the ratio κ of the mean model-based standard error to that standard deviation. Contrasts between weight cells of the same design and half-life are paired. Their MCSE is the standard deviation of the contrast over 2000 bootstrap resamples of the replicates. Contrasts across designs or half-lives combine the two cell MCSEs in quadrature.

4.6 Validation gate

For the first 50 replicates of every design and half-life, the driver regenerated the four single-channel boundary cells under the single-encoding DGPs from the same seeds. That gave 1,800 comparisons. The mean-shift encoding with weight a was compared with the cell $(a, 0)$, and

the correlation encoding with weight b with the cell $(0, b)$. The generated data were identical in every comparison, with a largest absolute difference of 0. The fitted E2 estimates were identical too, with a largest difference of 0. The gate passed. The boundary cells of the grid are the two single-encoding DGPs of this driver exactly. Of the 216,000 E2 and E2cr2 fits, 215,968 converged.

4.7 Test calibration under the null

Table 1 gives the rejection rate in the null cell, $c_{bm,A} = c_{bm,B} = 0$, with 1000 replicates per cell. The MCSE at the nominal level is 0.007. The model-based E2 test rejected at 0.060 to 0.071 in CO, 0.036 to 0.048 in Hybrid and 0.018 to 0.044 in OL+BDC. Its size therefore depends on the design. It is slightly liberal in CO at $t_{1/2} = 1$ week and conservative in OL+BDC at the same half-life. The calibration ratio κ of E2 ranged over 0.92 to 0.99 in CO, 1.01 to 1.14 in Hybrid and 1.05 to 1.22 in OL+BDC, across all weight cells. The CR2 test rejected at 0.038 to 0.061 in the null cell over all designs and half-lives. Its κ ranged over 0.94 to 1.02.

A design-dependent size means that E2 power is not comparable across designs without qualification. A conservative test loses power, and a liberal one gains it. Within one design and half-life, the size is the same for every weight cell, so contrasts between weight cells are unaffected. We therefore report design comparisons with both tests.

Table 1: Rejection rate in the null cell ($c_{bm,A} = c_{bm,B} = 0$), 1000 replicates per cell, $N = 70$, MCSE in parentheses. E2 uses the model-based standard error, E2cr2 the CR2 standard error with a Satterthwaite test. Half-lives in weeks.

	E2			E2cr2		
Design	$t_{1/2} = 0$	0.5	1	$t_{1/2} = 0$	0.5	1
CO	0.061 (0.008)	0.060 (0.008)	0.071 (0.008)	0.049 (0.007)	0.061 (0.008)	0.061 (0.008)
Hybrid	0.036 (0.006)	0.048 (0.007)	0.045 (0.007)	0.038 (0.006)	0.055 (0.007)	0.045 (0.007)
OL+BDC	0.044 (0.006)	0.028 (0.005)	0.018 (0.004)	0.057 (0.007)	0.060 (0.008)	0.047 (0.007)

4.8 Power over the weight grid

Table 2 gives E2 power for every cell and Table 3 gives E2cr2 power on the same trials. Power rose with each weight, holding the other fixed, in every design, half-life and test. There were 0 exceptions among all ordered pairs of cells. Monotonicity follows from Equation (5). Both weights add to the same coefficient with the same sign. It says nothing about whether the channels act independently.

At $t_{1/2} = 0$ the two single-channel cells at weight 0.45 had similar power in every design. Under E2 the values were 0.751 (MCSE 0.014) and 0.767 (MCSE 0.013) in CO, 0.894 (MCSE 0.010) and 0.911 (MCSE 0.009) in Hybrid, and 0.801 (MCSE 0.013) and 0.826 (MCSE 0.012) in OL+BDC, for the mean-shift and correlation channels in that order. At $t_{1/2} = 1$ week they separated in Hybrid and OL+BDC (Section 4.11).

Table 2: Power of the E2 test (model-based standard error) over the channel weights, $N = 70$, 1000 replicates per cell, MCSE in parentheses. Rows give $c_{bm,A}$ (mean-shift channel), columns $c_{bm,B}$ (correlation channel). The column $c_{bm,B} = 0$ is the mean-shift encoding and the row $c_{bm,A} = 0$ the correlation encoding. Half-lives in weeks.

		$c_{bm,B}$ (correlation channel)		
Design, $t_{1/2}$	$c_{bm,A}$	0	0.225	0.45
CO, 0	0.000	0.061 (0.008)	0.281 (0.014)	0.767 (0.013)
	0.225	0.279 (0.014)	0.755 (0.014)	0.985 (0.004)
	0.450	0.751 (0.014)	0.979 (0.005)	0.999 (0.001)
CO, 0.5	0.000	0.060 (0.008)	0.264 (0.014)	0.745 (0.014)
	0.225	0.265 (0.014)	0.732 (0.014)	0.970 (0.005)
	0.450	0.727 (0.014)	0.970 (0.005)	1.000 (0.000)
CO, 1	0.000	0.071 (0.008)	0.263 (0.014)	0.722 (0.014)
	0.225	0.273 (0.014)	0.718 (0.014)	0.964 (0.006)
	0.450	0.720 (0.014)	0.959 (0.006)	0.999 (0.001)
Hybrid, 0	0.000	0.036 (0.006)	0.352 (0.015)	0.911 (0.009)
	0.225	0.360 (0.015)	0.897 (0.010)	1.000 (0.000)
	0.450	0.894 (0.010)	1.000 (0.000)	1.000 (0.000)
Hybrid, 0.5	0.000	0.048 (0.007)	0.304 (0.015)	0.817 (0.012)
	0.225	0.351 (0.015)	0.846 (0.011)	0.996 (0.002)
	0.450	0.868 (0.011)	0.995 (0.002)	0.999 (0.001)
Hybrid, 1	0.000	0.045 (0.007)	0.222 (0.013)	0.723 (0.014)
	0.225	0.290 (0.014)	0.767 (0.013)	0.985 (0.004)
	0.450	0.818 (0.012)	0.987 (0.004)	1.000 (0.000)
OL+BDC, 0	0.000	0.044 (0.006)	0.269 (0.014)	0.826 (0.012)
	0.225	0.269 (0.014)	0.798 (0.013)	0.993 (0.003)
	0.450	0.801 (0.013)	0.990 (0.003)	1.000 (0.000)
OL+BDC, 0.5	0.000	0.028 (0.005)	0.195 (0.013)	0.711 (0.014)
	0.225	0.247 (0.014)	0.753 (0.014)	0.980 (0.004)
	0.450	0.803 (0.013)	0.981 (0.004)	0.999 (0.001)
OL+BDC, 1	0.000	0.018 (0.004)	0.150 (0.011)	0.559 (0.016)
	0.225	0.229 (0.013)	0.660 (0.015)	0.955 (0.007)
	0.450	0.740 (0.014)	0.973 (0.005)	0.997 (0.002)

Table 3: Power of the E2cr2 test (CR2 standard error, Satterthwaite test) on the same trials as Table 2. MCSE in parentheses.

		$c_{bm,B}$ (correlation channel)		
Design, $t_{1/2}$	$c_{bm,A}$	0	0.225	0.45
CO, 0	0.000	0.049 (0.007)	0.244 (0.014)	0.731 (0.014)
	0.225	0.235 (0.013)	0.694 (0.015)	0.970 (0.005)
	0.450	0.689 (0.015)	0.959 (0.006)	0.997 (0.002)
CO, 0.5	0.000	0.061 (0.008)	0.224 (0.013)	0.723 (0.014)
	0.225	0.225 (0.013)	0.691 (0.015)	0.951 (0.007)
	0.450	0.680 (0.015)	0.942 (0.007)	0.999 (0.001)
CO, 1	0.000	0.061 (0.008)	0.244 (0.014)	0.695 (0.015)
	0.225	0.248 (0.014)	0.666 (0.015)	0.952 (0.007)
	0.450	0.665 (0.015)	0.945 (0.007)	0.998 (0.001)
Hybrid, 0	0.000	0.038 (0.006)	0.377 (0.015)	0.916 (0.009)
	0.225	0.360 (0.015)	0.892 (0.010)	1.000 (0.000)
	0.450	0.881 (0.010)	0.999 (0.001)	1.000 (0.000)
Hybrid, 0.5	0.000	0.055 (0.007)	0.305 (0.015)	0.840 (0.012)
	0.225	0.349 (0.015)	0.834 (0.012)	0.996 (0.002)
	0.450	0.851 (0.011)	0.994 (0.002)	0.999 (0.001)
Hybrid, 1	0.000	0.045 (0.007)	0.243 (0.014)	0.741 (0.014)
	0.225	0.288 (0.014)	0.767 (0.013)	0.981 (0.004)
	0.450	0.801 (0.013)	0.981 (0.004)	1.000 (0.000)
OL+BDC, 0	0.000	0.057 (0.007)	0.298 (0.014)	0.877 (0.010)
	0.225	0.292 (0.014)	0.837 (0.012)	0.996 (0.002)
	0.450	0.824 (0.012)	0.994 (0.002)	1.000 (0.000)
OL+BDC, 0.5	0.000	0.060 (0.008)	0.265 (0.014)	0.774 (0.013)
	0.225	0.315 (0.015)	0.790 (0.013)	0.988 (0.003)
	0.450	0.822 (0.012)	0.987 (0.004)	1.000 (0.000)
OL+BDC, 1	0.000	0.047 (0.007)	0.226 (0.013)	0.688 (0.015)
	0.225	0.295 (0.014)	0.745 (0.014)	0.973 (0.005)
	0.450	0.808 (0.012)	0.978 (0.005)	0.999 (0.001)

4.9 The fitted coefficient

At $t_{1/2} = 0$ the fitted coefficient was additive in the two weights, replicate by replicate. The paired interaction contrast $\hat{\beta}(0.45, 0.45) - \hat{\beta}(0.45, 0) - \hat{\beta}(0, 0.45) + \hat{\beta}(0, 0)$ averaged 1.4e-11 or less in absolute value. The reason is exact. At $t_{1/2} = 0$ the mean shift is a linear combination of the fixed-effect columns $B_i D_{bc,it}$ and $D_{bc,it}$. Adding it changes neither the REML variance estimates nor anything but the interaction estimate. At $t_{1/2} > 0$ the contrast was at most 1.3e-04, against estimates near 0.23 in size.

The two weights did not have equal coefficients at $t_{1/2} > 0$. The correlation channel's mean estimate stayed within 4 percent of its nominal value in every design and half-life. The mean-

shift channel's estimate at weight 0.45 and $t_{1/2} = 1$ week exceeded the nominal -0.234 by 2 percent in CO, 15 percent in Hybrid and 20 percent in OL+BDC. The MCSE of each mean estimate was about 0.003 or less. The GLS projection under the true conditional covariance gave 1, 8 and 7 percent (Section 3). The fitted model weights the off-drug visits differently from the true covariance. Its overshoot was roughly twice the projection's in Hybrid and three times in OL+BDC. The 2026-10-28 internal review reported 21 to 39 percent for the earlier run, which lacked the baseline visit and used the original carryover rule. The direction agrees, and the size depends on the analysis data.

The empirical standard deviation of the E2 estimate exceeded the GLS standard error under the true covariance in every cell. The ratio of the two ranged from 1.08 to 1.26. The working covariance of the analysis model is a random intercept with continuous AR(1) errors. The DGP has four correlated components, so the fitted estimator is less efficient than GLS. The GLS noncentrality therefore overstates the attainable power and serves as an upper reference only.

4.10 Additivity of power

Table 4 compares the combined cell $(0.225, 0.225)$ with three benchmarks. The first is the probability-scale benchmark $p(0.225, 0) + p(0, 0.225) - p(0, 0)$. The second is the probit benchmark of the 2026-10-28 internal review. It maps each single-channel power to a noncentrality, $\hat{\Lambda} = \Phi^{-1}(\hat{\pi}) + 1.96$, adds the two and maps back, $\Phi(\hat{\Lambda}_A + \hat{\Lambda}_B - 1.96)$. The third is the same benchmark with the critical value calibrated to the observed null rejection rate, $z = \Phi^{-1}(1 - \hat{\pi}_0/2)$. It corrects for a test whose size is not 0.05. The table also gives the within-channel control. It compares $p(0.45, 0)$ with $2p(0.225, 0) - p(0, 0)$ for the mean-shift channel and the analogous quantity for the correlation channel.

The combined cell exceeded the probability-scale benchmark by 0.221 to 0.339 under E2. The single channels doubled showed the same excess, 0.210 to 0.337 for the mean-shift channel and 0.243 to 0.349 for the correlation channel. The excess is therefore a property of the power curve, not of combining two channels. At $(0.45, 0.45)$ the probability-scale benchmark exceeded 1 in every design and half-life. The combined power fell short of it by 0.284 to 0.769.

On the probit scale the combined cell agreed with the benchmark. Under E2 the departure was -0.047 to 0.028 in CO and Hybrid, within about two MCSE in every cell. In OL+BDC it was 0.031 to 0.092. These are the cells where E2 was conservative. With the critical value calibrated to the observed size, the OL+BDC departures were -0.062 to 0.015, and all nine departures were within two MCSE. Under the CR2 test, whose size was near nominal, the uncalibrated probit departures were -0.055 to 0.026. The probit approximation ignores the lower rejection tail and the t reference distribution, so agreement within a few hundredths is what it can deliver.

Table 4: Additivity of E2 power at the combined cell $(0.225, 0.225)$ and the within-channel control. Observed: power at $(0.225, 0.225)$. Excess over the probability-scale benchmark, over the probit benchmark with $z = 1.96$, and over the probit benchmark with z calibrated to the observed null rejection. Doubling: $p(0.45, 0) - [2p(0.225, 0) - p(0, 0)]$ for the mean-shift channel and the same for the correlation channel. Paired bootstrap MCSE in parentheses.

Design	$t_{1/2}$	Observed	Prob.	Probit	Probit, cal.	Doubling, mean	Doubling, cor.
CO	0	0.755	0.256 (0.026)	-0.031 (0.024)	-0.005 (0.027)	0.254 (0.026)	0.266 (0.026)
	0.5	0.732	0.263 (0.025)	-0.026 (0.024)	-0.001 (0.028)	0.257 (0.025)	0.277 (0.025)
	1	0.718	0.253 (0.024)	-0.047 (0.024)	0.003 (0.027)	0.245 (0.025)	0.267 (0.024)
Hybrid	0	0.897	0.221 (0.028)	0.008 (0.016)	-0.016 (0.015)	0.210 (0.028)	0.243 (0.029)
	0.5	0.846	0.239 (0.027)	-0.010 (0.018)	-0.014 (0.020)	0.214 (0.028)	0.257 (0.027)
	1	0.767	0.300 (0.025)	0.028 (0.026)	0.013 (0.029)	0.283 (0.028)	0.324 (0.025)
OL+BDC	0	0.798	0.304 (0.026)	0.031 (0.025)	0.015 (0.027)	0.307 (0.026)	0.333 (0.026)
	0.5	0.753	0.339 (0.024)	0.092 (0.030)	0.010 (0.032)	0.337 (0.026)	0.349 (0.025)
	1	0.660	0.299 (0.022)	0.088 (0.031)	-0.062 (0.038)	0.300 (0.025)	0.277 (0.022)

4.11 The equal-total line and carryover

The anti-diagonal of the grid holds three pairs with total weight 0.45: $(0.45, 0)$, $(0.225, 0.225)$ and $(0, 0.45)$. On drug they give the same conditional slope. They differ in the off-drug profile only. Table 5 gives their power and the paired difference between the two single-channel ends.

At $t_{1/2} = 0$ the correlation end had slightly higher power than the mean-shift end. The paired E2 differences, mean-shift minus correlation, were -0.025 to -0.016. This is the effect of the smaller conditional variance under the correlation channel. At $t_{1/2} = 1$ week the order reversed in Hybrid and OL+BDC. The differences were 0.095 (0.009) and 0.181 (0.012). In CO the difference stayed near zero, -0.002 (0.006). The mixed pair $(0.225, 0.225)$ lay between the two ends, or within two paired MCSE of the nearer end, in every design and half-life. Under E2cr2 the reversal at $t_{1/2} = 1$ week was 0.060 (0.008) in Hybrid and 0.120 (0.010) in OL+BDC. Pairs with the same total weight therefore do not give similar power once carryover is present. The earlier draft stated that they did.

The loss of power from $t_{1/2} = 0$ to 1 week shows the same difference. Under E2, the mean-shift encoding at weight 0.45 lost 0.076 in Hybrid and 0.061 in OL+BDC. The correlation encoding lost 0.188 and 0.267. The MCSE of each loss is about 0.020. Under E2cr2 the losses were 0.080 and 0.016 for the mean-shift encoding and 0.175 and 0.189 for the correlation encoding. Part of the E2 loss in OL+BDC reflects the test's growing conservatism there. In CO all losses were small, 0.031 to 0.045 under E2. The correlation channel's interaction decays with the drug effect. The mean-shift channel's interaction stops at discontinuation, so the off-drug visits keep their contrast with the on-drug visits. The earlier draft stated that the two channels contribute comparably to carryover sensitivity. In the Hybrid and OL+BDC designs the correlation channel lost more than twice as much power under either test.

Table 5: Power along the equal-total line $c_{bm,A} + c_{bm,B} = 0.45$, E2 and E2cr2, MCSE in parentheses. Difference: power at $(0.45, 0)$ minus power at $(0, 0.45)$, paired, with bootstrap MCSE. Half-lives in weeks.

Design	$t_{1/2}$	Test	$(0.45, 0)$	$(0.225, 0.225)$	$(0, 0.45)$	Difference
CO	0	E2	0.751 (0.014)	0.755 (0.014)	0.767 (0.013)	-0.016 (0.006)
		E2cr2	0.689 (0.015)	0.694 (0.015)	0.731 (0.014)	-0.042 (0.007)
	0.5	E2	0.727 (0.014)	0.732 (0.014)	0.745 (0.014)	-0.018 (0.007)
		E2cr2	0.680 (0.015)	0.691 (0.015)	0.723 (0.014)	-0.043 (0.007)
	1	E2	0.720 (0.014)	0.718 (0.014)	0.722 (0.014)	-0.002 (0.006)
		E2cr2	0.665 (0.015)	0.666 (0.015)	0.695 (0.015)	-0.030 (0.006)
Hybrid	0	E2	0.894 (0.010)	0.897 (0.010)	0.911 (0.009)	-0.017 (0.007)
		E2cr2	0.881 (0.010)	0.892 (0.010)	0.916 (0.009)	-0.035 (0.007)
	0.5	E2	0.868 (0.011)	0.846 (0.011)	0.817 (0.012)	0.051 (0.007)
		E2cr2	0.851 (0.011)	0.834 (0.012)	0.840 (0.012)	0.011 (0.007)
	1	E2	0.818 (0.012)	0.767 (0.013)	0.723 (0.014)	0.095 (0.009)
		E2cr2	0.801 (0.013)	0.767 (0.013)	0.741 (0.014)	0.060 (0.008)
OL+BDC	0	E2	0.801 (0.013)	0.798 (0.013)	0.826 (0.012)	-0.025 (0.009)
		E2cr2	0.824 (0.012)	0.837 (0.012)	0.877 (0.010)	-0.053 (0.009)
	0.5	E2	0.803 (0.013)	0.753 (0.014)	0.711 (0.014)	0.092 (0.010)
		E2cr2	0.822 (0.012)	0.790 (0.013)	0.774 (0.013)	0.048 (0.008)
	1	E2	0.740 (0.014)	0.660 (0.015)	0.559 (0.016)	0.181 (0.012)
		E2cr2	0.808 (0.012)	0.745 (0.014)	0.688 (0.015)	0.120 (0.010)

The ranking of designs depended on the channel and the half-life. The Hybrid design had the highest power, or power within two MCSE of the highest, in every single-channel cell at weight 0.45 under both tests. At weight 0.225 and $t_{1/2} = 1$ week, CO exceeded Hybrid on the correlation channel under E2, 0.041 (MCSE 0.019), but not under E2cr2, 0.001 (MCSE 0.019). That E2 difference reflects the liberal E2 test in CO. The order of CO and OL+BDC was not fixed. At $t_{1/2} = 0$ and weight 0.45 on the correlation channel, OL+BDC exceeded CO by 0.146 (MCSE 0.017) under E2cr2. At $t_{1/2} = 1$ week the difference was -0.007 (MCSE 0.021). On the mean-shift channel at $t_{1/2} = 1$ week, OL+BDC exceeded CO by 0.143 (MCSE 0.019). Under E2 the CO and OL+BDC differences also carry the two tests' different sizes. The earlier draft stated that the design ranking was preserved throughout the grid. For CO and OL+BDC it is not.

4.12 The G9 analysis and the pharmacological half-life

G9 chooses the analysis half-life by AIC over 0.25, 0.5, 1 and 2 weeks and tests with CR2[16]. Its cost is about eight E2 fits per trial, so we ran it on the first 200 replicates of four cells per design and half-life: the null cell and the three equal-total pairs. The trials are the same as those of the main run under the same seeds, so G9 and E2cr2 are paired. In the null cell G9 rejected at 0.035 to 0.085, against 0.030 to 0.085 for E2cr2 on the same trials. The MCSE at the nominal level is 0.015.

Table 6: G9 against E2cr2 on the same 200 trials per cell, $N = 70$, MCSE in parentheses. G9 chooses the analysis half-life by AIC. E2cr2 uses the true half-life. Mean $\widehat{t}^{a}_{1/2}$: the mean half-life G9 chose, in weeks.

Design	$t_{1/2}$	Cell	E2cr2	G9	Mean $\widehat{t}^{a}_{1/2}$
CO	0	$(0, 0)$	0.030 (0.012)	0.035 (0.013)	0.58
		$(0.45, 0)$	0.665 (0.033)	0.680 (0.033)	0.59
		$(0.225, 0.225)$	0.665 (0.033)	0.680 (0.033)	0.59
		$(0, 0.45)$	0.715 (0.032)	0.720 (0.032)	0.59
	0.5	$(0, 0)$	0.050 (0.015)	0.060 (0.017)	0.64
		$(0.45, 0)$	0.685 (0.033)	0.705 (0.032)	0.62
		$(0.225, 0.225)$	0.695 (0.033)	0.720 (0.032)	0.63
		$(0, 0.45)$	0.725 (0.032)	0.735 (0.031)	0.66
	1	$(0, 0)$	0.085 (0.020)	0.085 (0.020)	0.98
		$(0.45, 0)$	0.660 (0.033)	0.665 (0.033)	0.91
		$(0.225, 0.225)$	0.650 (0.034)	0.660 (0.033)	0.96
		$(0, 0.45)$	0.670 (0.033)	0.685 (0.033)	1.02
Hybrid	0	$(0, 0)$	0.035 (0.013)	0.040 (0.014)	0.29
		$(0.45, 0)$	0.885 (0.023)	0.885 (0.023)	0.29
		$(0.225, 0.225)$	0.895 (0.022)	0.899 (0.021)	0.29
		$(0, 0.45)$	0.915 (0.020)	0.915 (0.020)	0.28
	0.5	$(0, 0)$	0.045 (0.015)	0.055 (0.016)	0.68
		$(0.45, 0)$	0.825 (0.027)	0.835 (0.026)	0.53
		$(0.225, 0.225)$	0.820 (0.027)	0.825 (0.027)	0.58
		$(0, 0.45)$	0.810 (0.028)	0.815 (0.027)	0.63
	1	$(0, 0)$	0.080 (0.019)	0.080 (0.019)	1.47
		$(0.45, 0)$	0.810 (0.028)	0.805 (0.028)	1.13
		$(0.225, 0.225)$	0.775 (0.030)	0.765 (0.030)	1.27
		$(0, 0.45)$	0.770 (0.030)	0.770 (0.030)	1.43
OL+BDC	0	$(0, 0)$	0.065 (0.017)	0.065 (0.017)	0.28
		$(0.45, 0)$	0.830 (0.027)	0.825 (0.027)	0.27
		$(0.225, 0.225)$	0.845 (0.026)	0.835 (0.026)	0.27
		$(0, 0.45)$	0.865 (0.024)	0.865 (0.024)	0.27
	0.5	$(0, 0)$	0.045 (0.015)	0.045 (0.015)	0.46
		$(0.45, 0)$	0.840 (0.026)	0.845 (0.026)	0.44
		$(0.225, 0.225)$	0.810 (0.028)	0.810 (0.028)	0.46
		$(0, 0.45)$	0.780 (0.029)	0.780 (0.029)	0.48
	1	$(0, 0)$	0.045 (0.015)	0.040 (0.014)	0.77
		$(0.45, 0)$	0.815 (0.027)	0.820 (0.027)	0.68
		$(0.225, 0.225)$	0.745 (0.031)	0.750 (0.031)	0.71
		$(0, 0.45)$	0.680 (0.033)	0.675 (0.033)	0.77

At $t_{1/2} = 1$ week in OL+BDC, G9 gave 0.820 (0.027) at the mean-shift end and 0.675 (0.033) at the correlation end of the equal-total line. E2cr2 gave 0.815 (0.027) and 0.680 (0.033) on the same trials. In Hybrid, G9 gave 0.805 (0.028) and 0.770 (0.030), and E2cr2 gave 0.810 (0.028) and 0.770 (0.030). Over all 36 cells, G9 power differed from E2cr2 power on the same trials by at most 0.025. Choosing the half-life by AIC therefore cost little power here, and

the difference between the two ends of the equal-total line remained under G9. G9 cannot select $t^a_{1/2} = 0$, since its grid starts at 0.25 week. At $t_{1/2} = 0$ it chose a mean half-life of 0.27 to 0.59 week.

The pharmacological range of the prazosin half-life is 0.1 to 0.2 week. We ran the full weight grid at $t_{1/2} = 0.2$ week with 1000 replicates per cell, as a separate pass of the same driver. Its validation gate passed in 600 further comparisons. Under E2 the null cell rejected at 0.056 (MCSE 0.007) in CO, 0.050 (MCSE 0.007) in Hybrid and 0.026 (MCSE 0.005) in OL+BDC. At weight 0.45, the mean-shift and correlation encodings had power 0.734 (MCSE 0.014) and 0.743 (MCSE 0.014) in CO, 0.882 (MCSE 0.010) and 0.895 (MCSE 0.010) in Hybrid, and 0.810 (MCSE 0.012) and 0.821 (MCSE 0.012) in OL+BDC. The paired differences were -0.009 (0.006), -0.013 (0.007) and -0.011 (0.009). The GLS projection puts the mean-shift overshoot at 0 to 1 percent at this half-life. At pharmacological half-lives the two channels are therefore close to interchangeable, and the dual-channel architecture adds little to a single encoding.

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5 When to use the dual-channel architecture

The dual-channel DGP does not represent two biological mechanisms. Within it, the two weights describe one conditional-mean slope and how much of that slope persists after discontinuation. The mean-shift channel's interaction stops at discontinuation. The correlation channel's interaction decays at the carryover half-life. The ratio of the weights fixes the off-drug profile of the interaction.

This gives the architecture a specific use. A simulation study usually does not know whether the biomarker's influence on response outlasts the drug. A sweep along the equal-total line, $\omega \in [0, 1]$ at a fixed total weight, shows how power depends on that unknown profile. It holds the on-drug interaction fixed. At $t_{1/2} = 0$ the sweep is nearly flat, so it is informative only when carryover is plausible.

The architecture is not a model of a drug that changes the spread of response in high-biomarker participants. That question concerns the second moment. It needs a variance function[8] or a responder mixture[7]. In the prazosin-PTSD example, the question is whether blood pressure predicts a larger prazosin effect, and whether that predictive association fades after discontinuation at the rate the drug effect does. The dual-channel architecture can represent both answers to the second question. It cannot represent heterogeneity of response variance.

As a default for a single power calculation, we recommend one of the two encodings, with its weight stated. The correlation encoding is the cautious choice when carryover is plausible. Its interaction decays with the drug effect. In the Hybrid and OL+BDC designs its power loss under carryover was more than twice that of the mean-shift encoding (Section 4.11). A second free weight adds nothing to a power calculation unless the off-drug profile is itself in question.

6 Discussion

The dual-channel DGP nests the two encodings of the biomarker-treatment interaction. Under a normally distributed biomarker it is one Gaussian model. Both weights set the conditional-mean slope of the drug-response component on the biomarker. They differ in two respects only. The correlation weight also lowers the on-drug residual variance, by about 2 percent at the outcome level in this calibration. The two weights also give different off-drug profiles. These facts replace the earlier draft's account of two mechanisms acting on two moments.

The fitted interaction coefficient is linear in the two weights. Its coefficients are equal when there is no carryover and differ when there is. The difference comes from the mean-shift channel. Its interaction stops at discontinuation, but the analysis regressor $D_{bc,it}$ decays. The projection of its step-shaped profile onto the regressor overshoots the nominal value. The correlation channel's profile matches the regressor at the true half-life and gives the nominal value exactly.

Power is not additive in the weights, but nothing about the channels causes that. Additivity holds for the coefficient and, to a close approximation, for the noncentrality. Excess over the probability-scale benchmark is the curvature of the power function. A single channel with its weight doubled shows the same excess. The title of the earlier draft named a "channel super-additivity". We withdraw that term.

The earlier run failed its own validation. Its boundary cells fell 10 to 19 points below the single-encoding power reported in an earlier version of paper 01. The earlier draft attributed the gap to mismatched calibration inputs. Those inputs were in fact identical in the two drivers. The 2026-10-28 internal review traced most of the gap to the analysis data. The package analysis that produced paper 01's numbers adds the baseline visit. The combined driver did not. Adding the baseline visit closed the gap in CO and OL+BDC to within one MCSE in a 400-replicate check. In the Hybrid design a gap of 0.03 to 0.06 remained, 1.3 to 2.5 MCSE. We have not located its source. One candidate is a difference between the package data generator and the `implementations/tidyverse` generator the combined driver uses. We do not compare with paper 01's numbers here. They come from a version of paper 01 that has since been restated, under the earlier DGP. The validation gate of Section 4.6 instead compares each boundary cell with the single-encoding DGP of the same driver, draw for draw.

The study has limitations. The half-lives 0.5 and 1 week are behavioral scenarios. At 0.2 week, inside the pharmacological range of prazosin, the two channels were close to interchangeable, so the differences reported here matter mainly for behavioral carryover. The G9 arm covered 200 replicates of four cells per design and half-life, so its comparisons are coarser than the main grid. The biomarker is normal, so the two channels share the conditional mean by

construction. A skewed or discrete biomarker would break the equivalence. We did not fit a specification that models the off-drug slope separately, so we have not shown that the off-drug profile is estimable in practice. The study is exploratory.

Glossary

- **Dual-channel architecture.** The DGP of this paper, with both channel weights free. Data value `combined`. Formerly "Architecture C".
- **Mean-shift encoding, mean-shift channel.** The interaction as a shift $c_{bm,A}\, b_i\, \sigma_{BR}$ of the drug-response component at on-drug visits. Data value `mean_moderation`. NOTATION.md calls it the mean-moderation architecture.
- **Correlation encoding, correlation channel.** The interaction as a correlation $c_{bm,B}\, g_t$ between the biomarker and the drug-response component, present on drug and decaying off drug. Data value `mvn`. NOTATION.md calls it the covariance-moderation architecture.
- $c_{bm,A}$, $c_{bm,B}$. The mean-shift and correlation weights (code `c.bm_a`, `c.bm_b`). The subscripts are symbol labels only.
- ω. Share of a fixed total weight assigned to the correlation channel.
- g_t. Correlation profile: the fraction of $c_{bm,B}$ present at visit t.
- $D_{bc,it}$. Exposure-decayed drug indicator of the analysis model (code `Dbc`).
- κ_A, κ_B. Coefficients of the two weights in the population value of $\beta_{bm:D}$ (Equation (5)).
- κ. Calibration ratio: mean model-based standard error divided by the empirical standard deviation of the estimates. Values above 1 mean a conservative test.
- Λ. Noncentrality of the interaction test.
- **Extended DGP.** The DGP of the companion paper[2]: decaying biomarker-response correlation (0 before first exposure), the same AR(1) correlation for every component with a constant between-component correlation at every lag, carryover that decays from the last on-drug mean, and a stop on a matrix that is not positive definite. Code label `refined`.
- **Original carryover rule.** The drug-response mean at an off-drug visit is the previous visit's mean times the decay factor. The earlier run of this paper used it.
- **Largest feasible correlation.** The largest $c_{bm,B}$ for which every path's covariance matrix is positive definite.
- **E2, E2cr2, G8, G9.** Analysis codes. E2 is the exposure-weighted interaction model with a model-based test. E2cr2 (G8) is the same fit with a CR2 test. G9 chooses the half-life by AIC and uses a CR2 test.
- **CO.** Two-period crossover.
- **OL+BDC.** Open-label titration followed by blinded discontinuation.
- **Hybrid.** Open-label titration, blinded discontinuation, then a brief crossover.

- **Common random numbers.** The same random deviates used for every weight cell of a replicate.
- **Probability-scale benchmark.** $p(a,0) + p(0,b) - p(0,0)$.
- **Probit benchmark.** $\Phi(\hat{\Lambda}_a + \hat{\Lambda}_b - z)$ with $\hat{\Lambda} = \Phi^{-1}(\hat{\pi}) + z$, where z is the critical value.

Abbreviations

AIC, Akaike information criterion. AR(1), first-order autoregressive. CR2, bias-reduced cluster-robust variance estimator. DGP, data-generating process. GLS, generalized least squares. MCSE, Monte Carlo standard error. PTSD, posttraumatic stress disorder. REML, restricted maximum likelihood.

Conflict of interest

The pmsimstats team declares no conflicts of interest.

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Reproducibility

The driver is `analysis/scripts/dgp-combined/01-run-combined-factorial.R`, run through `analysis/scripts/dgp-combined/run-combined.sh`, which sets the BLAS thread variables to 1. Its header lists the exact invocations. It sources the data generator in `implementations/tidyverse/R/functions.R` and the shared helpers in `analysis/scripts/carryover-sensitivity/simulation-core.R`. The seed of each replicate is a hash of the design, half-life and replicate index (`label_seed()`). Results do not depend on the number of workers. `02-summarise-combined.R` produces the summaries, CSV files and bootstrap MCSEs. `03-analytic-projection.R` produces the closed-form quantities. Outputs are in `analysis/data/dgp-combined/`. The main grid has untagged file names. The $t_{1/2} = 0.2$ week pass and the G9 pass carry the suffixes `-hl02` and `-g9` and have their own `meta` files. The stored run was made on 2026-10-29 12:36 PDT with R version 4.6.1 (2026-06-24) on aarch64-apple-darwin25.4.0, nlme 3.1.169 and clubSandwich 0.7.0, from commit 8918e5c6e. The driver and its sourced files had uncommitted changes at the time, so the commit alone does not reproduce the run. It took 168 minutes of wall time on 8 workers. It ran on the host, not in the project container. Results from the container and from the host are not identical draw for draw, because the BLAS and LAPACK libraries differ.

The run of 2026-07-31 that the earlier draft reported is archived in `analysis/data/dgp-combined/archive-202`
It used the original carryover rule, a cross-component correlation of 0.1 at nonzero lags, an

exposure indicator without the pre-exposure guard, repair of matrices that were not positive definite, no baseline visit in the analysis data and independent random streams per cell. Under the missing guard, placebo-first CO participants were coded as exposed at every visit when $t_{1/2} > 0$. No matrix in that run needed repair. The manuscript source, bibliography and CSL file are in `analysis/report/11-combined-dgp-architecture/`.

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